CS5340 - Uncertainty Modelling in AI

Lecture 8: Hidden Markov Models (HMM)

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Semester 1, AY2026/27 — National University of Singapore

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1 Lecture Overview

- **Motivation:** in Lecture 7 the latent "which component" variable $\mathbf{z}_n$ of each observation was chosen **independently**. Many signals – speech, trajectories, prices – are **sequential**: which component generated $\mathbf{x}_n$ depends on which one generated $\mathbf{x}_{n-1}$. The **hidden Markov model** makes the latent variables a **Markov chain**.
- **What carries over:** the HMM is still a latent-variable model, so its parameters are learned with **EM** (Lecture 7), and its posteriors are computed by **message passing on a chain** (Lectures 4–5).
- **Roadmap:**
  1. **The model**: transition probabilities $\mathbf{A}$, initial distribution $\boldsymbol{\pi}$, emission probabilities $\boldsymbol{\phi}$, and the joint $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$.
  2. **EM for HMMs**: the expected complete-data log-likelihood needs only the single-node posteriors $\gamma(\mathbf{z}_n)$ and the pairwise posteriors $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$; the M-step is closed-form.
  3. **The forward-backward (α-β) algorithm**: computes γ and ξ in $O(K^2N)$ instead of $O(K^N)$; it is exactly **sum-product** on the HMM's factor graph. Used inside EM it is the **Baum-Welch** algorithm.
  4. **Scaling factors**: a normalized recursion that avoids numerical underflow.
  5. **The Viterbi algorithm**: **max-sum** on the chain plus back-tracking, giving the most probable **sequence** of hidden states.

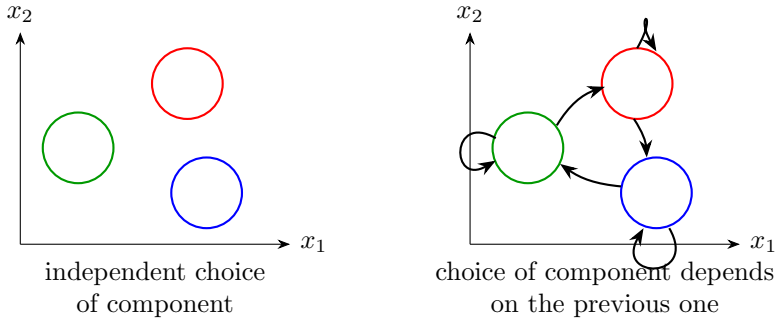
Learning Outcomes

By the end of this lecture, you should be able to:

1. Describe the joint distribution of a HMM with the **transition and emission probabilities**.
2. Use the **EM algorithm** for maximum likelihood estimation of the latent variables and unknown parameters in the HMM.
3. Use the **forward-backward algorithm** to evaluate the EM algorithm.
4. Apply the **Viterbi algorithm** to find the maximal probability and its configuration.

2 From Mixture Models to Sequential Data

In a mixture model (Lecture 7) the choice of mixture component for each observation is **independent**: every $\mathbf{x}_n$ draws its own $\mathbf{z}_n \sim \text{Cat}(\boldsymbol{\pi})$. An HMM is an **extension** of the mixture model in which the choice of component **depends on the choice of component for the previous observation** – e.g. the data tend to stay in one cluster for a while, then jump to another along preferred routes.

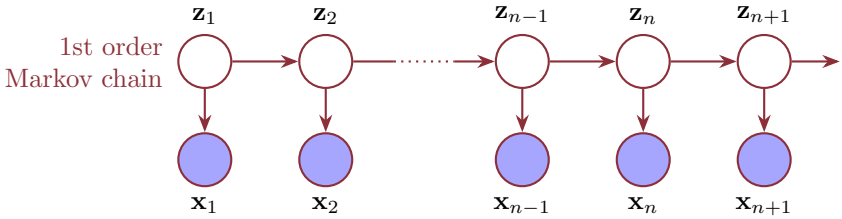


(Schematic redrawn by the scribes after Jordan, 2002.)

Definition : Hidden Markov Model (HMM)

- An HMM is a natural generalization of a mixture model – a **"dynamic" mixture model**.
- For each **observed variable** X_n there is a corresponding **latent variable** Z_n (which may be of a different type or dimensionality to X_n).
- The latent variables $\{Z_1, \ldots, Z_{n-1}, Z_n, \ldots\}$ form a **Markov chain**: the current state Z_n depends on the previous state Z_{n-1}. This structure is a **state space model**.
- Key **conditional independence**: Z_{n-1} and Z_{n+1} are independent given Z_n,

$$\mathbf{z}_{n+1} \perp\!\!\!\perp \mathbf{z}_{n-1} \mid \mathbf{z}_n.$$



The HMM as a DGM: hidden $\mathbf{z}_n$ (white), observed $\mathbf{x}_n$ (shaded). *(Redrawn after Bishop.)*

Definition : HMM Joint Distribution

Reading the factorization off the DGM (Lecture 2),

$$p(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{z}_1, \ldots, \mathbf{z}_N) = p(\mathbf{z}_1) \left[\prod_{n=2}^{N} p(\mathbf{z}_n \mid \mathbf{z}_{n-1}) \right] \prod_{n=1}^{N} p(\mathbf{x}_n \mid \mathbf{z}_n).$$

- **HMM** (this lecture): latent variables **must be discrete**; observed variables can be either discrete or continuous.
- **Linear dynamical system** (not covered in this course): latent and observed variables are both **continuous**; **linear-Gaussian** if both are Gaussian (e.g. the Kalman filter).

Worked Example: Where HMMs Are Used

- **Speech recognition:** given an audio waveform (and its spectrogram), robustly extract and recognize the spoken words – the hidden states are phonemes (e.g. "b ey z th ih er em" for "Bayes' Theorem"), the observations are acoustic features. (Image source: Bishop.)
- **Target tracking:** estimate the motion of targets in the 3D world from indirect, potentially noisy measurements (e.g. a ship's tracking beam following a crossing target).
- **Robotic SLAM:** simultaneous localization and mapping – pose and world geometry estimated as the robot moves; the robot poses form the chain, with controls u_k and landmark observations z_k of map features m_i. (S. Thrun, San Jose Tech Museum.)
- **Financial forecasting:** predict future market behaviour (e.g. the Straits Times Index) from historical data, with hidden "market regimes".

Scribe's note: target tracking and SLAM have continuous states, so strictly they are state space models of the linear-dynamical-system kind (or nonlinear generalizations); they share the HMM's chain structure and the same forward-backward logic.

3 Transition and Emission Probabilities

3.1 Transition Probabilities

Definition : Transition Matrix A

The discrete latent variables use the **1-of-K coding scheme** $\mathbf{z}_n \in \{0,1\}^K$ (Lecture 7), describing **which component of the mixture is responsible for generating** $\mathbf{x}_n$. $\mathbf{z}_n$ depends on $\mathbf{z}_{n-1}$ through a conditional distribution $p(\mathbf{z}_n \mid \mathbf{z}_{n-1})$, which corresponds to a $K \times K$ **state transition matrix** $\mathbf{A} \in \mathbb{R}^{K \times K}$ whose elements are the **transition probabilities**:

1. $A_{jk} \equiv p(z_{nk} = 1 \mid z_{n-1,j} = 1)$ (row j = **from**, column k = **to**);
2. $0 \le A_{jk} \le 1$, with $\sum_k A_{jk} = 1$ (each **row** sums to one);
3. $K(K-1)$ **independent parameters**.

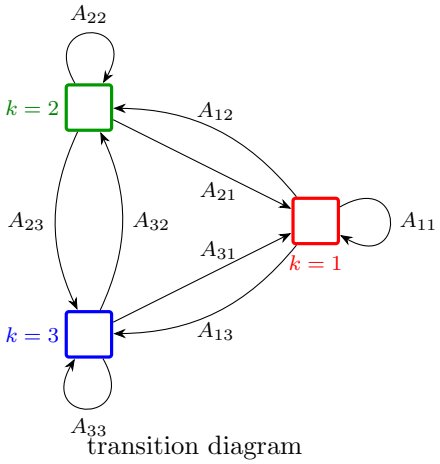
Explicitly,

$$p(\mathbf{z}_n \mid \mathbf{z}_{n-1}, \mathbf{A}) = \prod_{k=1}^{K} \prod_{j=1}^{K} A_{jk}^{z_{n-1,j} z_{nk}}.$$

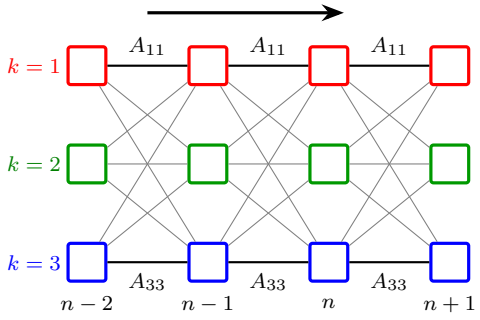
The **initial** latent variable $\mathbf{z}_1$ has no parent, so it is a **categorical distribution**

$$p(\mathbf{z}_1 \mid \boldsymbol{\pi}) = \prod_{k=1}^{K} \pi_k^{z_{1k}}, \qquad \sum_k \pi_k = 1.$$

Only one factor in the double product survives: if $z_{n-1,j} = 1$ and $z_{nk} = 1$, every exponent except that of A_{jk} is 0 – the same trick as $p(\mathbf{z}) = \prod_k \pi_k^{z_k}$ in Lecture 7.



transition diagram



lattice / trellis (diagram unfolded in time)

(Redrawn after Bishop.) Each column of the trellis is one latent variable Z_n.

Key Difficulty: These Are NOT Graphical Models!

The **transition diagram** (nodes = the K **states** of one variable, edges = nonzero A_{jk}) and the **lattice/trellis** (the diagram unfolded over time, one column per Z_n) are **not graphical models**: their nodes are **values** of a variable, not random variables. The graphical model is the chain $\mathbf{z}_1 \to \mathbf{z}_2 \to \cdots$ with one node per Z_n. The trellis is nonetheless the right picture for the forward recursion and for Viterbi, since a path through it is one configuration $(\mathbf{z}_1, \ldots, \mathbf{z}_N)$.

3.2 Emission Probabilities

Definition : Emission Probabilities

The **emission probabilities** are the conditional distributions of the observed $\mathbf{x}_n$ given the latent $\mathbf{z}_n$, $p(\mathbf{x}_n \mid \mathbf{z}_n, \boldsymbol{\phi})$, where $\boldsymbol{\phi} = \{\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_K\}$ is a set of parameters governing the distribution:

$$p(\mathbf{x}_n \mid \mathbf{z}_n, \boldsymbol{\phi}) = \prod_{k=1}^{K} p(\mathbf{x}_n \mid \boldsymbol{\phi}_k)^{z_{nk}}.$$

- They can be **Gaussians** if $\mathbf{x}_n$ is continuous, or **conditional probability tables** if $\mathbf{x}_n$ is discrete.
- For a given (observed) $\mathbf{x}_n$ and a given $\boldsymbol{\phi}$, $p(\mathbf{x}_n \mid \mathbf{z}_n, \boldsymbol{\phi})$ is just a **vector of K numbers**, one for each possible state of $\mathbf{z}_n$.

Definition : Homogeneous Model and the Full Joint

In a **homogeneous** model, all the conditionals governing the latent variables share the **same $\mathbf{A}$**, and all the emission distributions share the **same $\boldsymbol{\phi}$**. The joint over latent and observed variables is then

$$p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = p(\mathbf{z}_1 \mid \boldsymbol{\pi}) \underbrace{\left[\prod_{n=2}^{N} p(\mathbf{z}_n \mid \mathbf{z}_{n-1}, \mathbf{A})\right]}_{\text{transition probabilities}} \underbrace{\prod_{m=1}^{N} p(\mathbf{x}_m \mid \mathbf{z}_m, \boldsymbol{\phi})}_{\text{emission probabilities}},$$

with $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, $\mathbf{Z} = \{\mathbf{z}_1, \ldots, \mathbf{z}_N\}$ and parameters $\boldsymbol{\theta} = \{\boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\phi}\}$.

Key Rule: The Mixture Model Is a Special Case

If all **rows of $\mathbf{A}$ are the same**, then $p(\mathbf{z}_n \mid \mathbf{z}_{n-1})$ does not depend on $\mathbf{z}_{n-1}$, the chain breaks into independent draws, and we recover the **mixture model for an i.i.d. data set** of Lecture 7 (with mixing coefficients equal to that common row).

Worked Example: A Running Example – Two Biased Coins (Scribe's Example)

Not from the slides; used throughout this scribe so every formula can be checked by hand. A dealer holds two coins. State $k = 1$: coin 1, which shows **H** with probability 0.8. State $k = 2$: coin 2, which shows **T** with probability 0.8. Before each toss the dealer keeps the same coin with probability 0.7. So $K = 2$ and

$$\boldsymbol{\pi} = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{bmatrix},$$

	$p(\mathrm{H} \mid k)$	$p(\mathrm{T} \mid k)$
$k = 1$	0.8	0.2
$k = 2$	0.2	0.8

We observe $N = 3$ tosses: $\mathbf{X} = (\mathrm{H}, \mathrm{T}, \mathrm{H})$. The emission vectors are therefore $p(\mathbf{x}_1 \mid \mathbf{z}_1) = p(\mathbf{x}_3 \mid \mathbf{z}_3) = (0.8, 0.2)$ and $p(\mathbf{x}_2 \mid \mathbf{z}_2) = (0.2, 0.8)$. There are $K^N = 8$ hidden paths – few enough to check everything by brute force.

4 Maximum Likelihood and EM for HMMs

Key Difficulty: The Likelihood Is a Sum over K^N Paths

The likelihood is obtained by **marginalizing over the latent variables**:

$$p(\mathbf{X} \mid \boldsymbol{\theta}) = \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}).$$

- We **cannot** treat each sum over $\mathbf{z}_n$ independently, because $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$ **does not factorize** over $\mathbf{z}_1, \ldots, \mathbf{z}_N$ (the transition factors couple neighbours).
- Performing the sums over all N variables naively has **exponential complexity** $O(K^N)$.
- Moreover, as with GMMs, direct maximization leads to **no closed-form solution** (sum inside the log).

4.1 The E-Step: What Must Be Computed

We turn to the **EM algorithm** (Lecture 7). Initialize the parameters $\boldsymbol{\theta}^{\text{old}}$. In the **E-step**, find the posterior of the latent variables $p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$ and use it to evaluate the expected complete-data log-likelihood

$$\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \sum_{\mathbf{Z}} p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}).$$

Definition : Single and Pairwise Posterior Marginals

$$\gamma(\mathbf{z}_n) = p(\mathbf{z}_n \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}), \qquad \xi(\mathbf{z}_{n-1}, \mathbf{z}_n) = p(\mathbf{z}_{n-1}, \mathbf{z}_n \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}).$$

- For each n, $\gamma(\mathbf{z}_n)$ is stored as K **nonnegative numbers that sum to unity**.
- $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$ is stored as a $K \times K$ **matrix of nonnegative numbers that sum to unity**.
- $\gamma(z_{nk})$ denotes the conditional probability that $z_{nk} = 1$; similarly $\xi(z_{n-1,j}, z_{nk})$ that $z_{n-1,j} = z_{nk} = 1$.

Proof: Expectations of Binary Indicators

The expectation of a binary random variable is the probability that it equals 1:

$$\gamma(z_{nk}) = \mathbb{E}[z_{nk}] = \sum_{\mathbf{z}_n} \gamma(\mathbf{z}_n)\, z_{nk},$$

$$\xi(z_{n-1,j}, z_{nk}) = \mathbb{E}[z_{n-1,j} z_{nk}] = \sum_{\mathbf{z}_{n-1}, \mathbf{z}_n} \xi(\mathbf{z}_{n-1}, \mathbf{z}_n)\, z_{n-1,j} z_{nk}.$$

In the first sum $\mathbf{z}_n$ runs over its K one-hot states; the indicator z_{nk} is 0 in every state except "$z_{nk} = 1$", so only the term $\gamma(z_{nk} = 1) \cdot 1$ survives. In the second sum $(\mathbf{z}_{n-1}, \mathbf{z}_n)$ runs over K^2 pairs of states, and $z_{n-1,j} z_{nk}$ is 1 only for the pair "$z_{n-1,j} = 1,\ z_{nk} = 1$", so only $\xi(z_{n-1,j}, z_{nk})$ survives. ■

Worked Example: Deriving $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$ for the HMM

Take the log of the joint, using the 1-of-K forms of each factor:

$$\ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} z_{1k} \ln \pi_k + \sum_{n=2}^{N} \sum_{k=1}^{K} \sum_{j=1}^{K} z_{n-1,j} z_{nk} \ln A_{jk} + \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} \ln p(\mathbf{x}_n \mid \boldsymbol{\phi}_k).$$

This is **linear** in the indicators z_{1k}, $z_{n-1,j} z_{nk}$ and z_{nk}. Taking the expectation under $p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$ therefore just replaces each indicator by its expectation, and each expectation only involves the posterior of the one or two variables it mentions:

$$\boxed{\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \sum_{k=1}^{K} \gamma(z_{1k}) \ln \pi_k + \sum_{n=2}^{N} \sum_{j=1}^{K} \sum_{k=1}^{K} \xi(z_{n-1,j}, z_{nk}) \ln A_{jk} + \sum_{n=1}^{N} \sum_{k=1}^{K} \gamma(z_{nk}) \ln p(\mathbf{x}_n \mid \boldsymbol{\phi}_k).}$$

So the E-step does **not** need the full posterior over K^N paths – only the N vectors $\gamma(\mathbf{z}_n)$ and the $N-1$ matrices $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$. The goal of the E-step is to evaluate these **efficiently**: that is the forward-backward algorithm (Section 5).

4.2 The M-Step

In the **M-step** we maximize $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$ w.r.t. $\boldsymbol{\theta} = \{\boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\phi}\}$, treating $\gamma(\mathbf{z}_n)$ and $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$ as **constants**. The three terms of $\mathcal{Q}$ involve disjoint parameters, so they are maximized separately.

Key Rule: M-Step Updates for π and A

Maximization w.r.t. $\boldsymbol{\pi}$ and $\mathbf{A}$ is achieved with **Lagrange multipliers** (constraints $\sum_k \pi_k = 1$ and $\sum_k A_{jk} = 1$ for each row j):

$$\boxed{\pi_k = \frac{\gamma(z_{1k})}{\sum_{j=1}^{K} \gamma(z_{1j})}}, \qquad \boxed{A_{jk} = \frac{\sum_{n=2}^{N} \xi(z_{n-1,j}, z_{nk})}{\sum_{l=1}^{K} \sum_{n=2}^{N} \xi(z_{n-1,j}, z_{nl})}}.$$

Derivation (scribe's addition). For row j of $\mathbf{A}$, maximize $\sum_n \sum_k \xi(z_{n-1,j}, z_{nk}) \ln A_{jk} + \lambda_j \big(\sum_k A_{jk} - 1\big)$. Setting the derivative w.r.t. A_{jk} to zero gives $\sum_n \xi(z_{n-1,j}, z_{nk}) / A_{jk} + \lambda_j = 0$, i.e. $A_{jk} \propto \sum_n \xi(z_{n-1,j}, z_{nk})$; normalizing over k gives the boxed result. $\boldsymbol{\pi}$ is identical with $\gamma(z_{1k})$ in place of ξ (and since $\sum_j \gamma(z_{1j}) = 1$, simply $\pi_k = \gamma(z_{1k})$).
Interpretation: A_{jk} = (expected number of $j \to k$ transitions) / (expected number of transitions out of j) – the Lecture 6 categorical MLE with **soft counts**.

Key Rule: M-Step Updates for the Emission Parameters ϕ

Gaussian emissions, $p(\mathbf{x}_n \mid \boldsymbol{\phi}_k) = \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$: the last term of $\mathcal{Q}$ is exactly the GMM objective of Lecture 7, so

$$\boldsymbol{\mu}_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk}) \mathbf{x}_n}{\sum_{n=1}^{N} \gamma(z_{nk})}, \qquad \boldsymbol{\Sigma}_k = \frac{\sum_{n=1}^{N} \gamma(z_{nk}) (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^T}{\sum_{n=1}^{N} \gamma(z_{nk})}.$$

Discrete (multinomial) emissions, $\mathbf{x}_n \in \{0,1\}^D$ with $\sum_{i=1}^{D} x_{ni} = 1$ (1-of-D coding):

$$p(\mathbf{x} \mid \mathbf{z}) = \prod_{i=1}^{D}\prod_{k=1}^{K} \mu_{ik}^{x_i z_k} \qquad \Longrightarrow \qquad \mu_{ik} = \frac{\sum_{n=1}^{N} \gamma(z_{nk})\, x_{ni}}{\sum_{n=1}^{N} \gamma(z_{nk})},$$

the expected fraction of times state k emits symbol i.

Key Difficulty: Initialization

1. EM must be **initialized** by choosing starting values for $\boldsymbol{\pi}$ and $\mathbf{A}$ that satisfy the **summation constraints**.
2. Initial values for $\boldsymbol{\pi}$ and **A cannot be set to zero**: they will remain zero in subsequent EM updates. (*Why:* $\xi(z_{n-1,j}, z_{nk})$ contains the factor A_{jk}^{old} – see Section 5 – so $A_{jk}^{\text{old}} = 0$ gives $\xi = 0$ and hence $A_{jk}^{\text{new}} = 0$; likewise for π_k through $\alpha(\mathbf{z}_1)$.)
3. Typically, we set **random starting values** satisfying the summation and non-negativity constraints.

Scribe's note: the "stuck at zero" property is sometimes used deliberately – e.g. setting $A_{jk} = 0$ for $k < j$ gives a left-to-right HMM (common in speech) that EM will preserve.

5 The Forward-Backward Algorithm

The **forward-backward algorithm** computes $\gamma(\mathbf{z}_n)$ and $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$ efficiently. Of its many variants we focus on the most widely used, the **alpha-beta algorithm**. When used inside EM for parameter learning, the whole procedure is known as the **Baum-Welch algorithm**.

Definition : Conditional Independence Properties of the HMM

Using **d-separation** on the HMM's DGM (every path between past and future passes through the unobserved $\mathbf{z}_n$ head-to-tail), with $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$:

$$
\begin{aligned}
p(\mathbf{X} \mid \mathbf{z}_n) &= p(\mathbf{x}_1, \ldots, \mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n) \\
p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{x}_n, \mathbf{z}_n) &= p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{z}_n) \\
p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{z}_{n-1}, \mathbf{z}_n) &= p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{z}_{n-1}) \\
p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n, \mathbf{z}_{n+1}) &= p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_{n+1}) \\
p(\mathbf{x}_{n+2}, \ldots, \mathbf{x}_N \mid \mathbf{z}_{n+1}, \mathbf{x}_{n+1}) &= p(\mathbf{x}_{n+2}, \ldots, \mathbf{x}_N \mid \mathbf{z}_{n+1}) \\
p(\mathbf{X} \mid \mathbf{z}_{n-1}, \mathbf{z}_n) &= p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{z}_{n-1})\, p(\mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n) \\
p(\mathbf{x}_{N+1} \mid \mathbf{X}, \mathbf{z}_{N+1}) &= p(\mathbf{x}_{N+1} \mid \mathbf{z}_{N+1}) \\
p(\mathbf{z}_{N+1} \mid \mathbf{z}_N, \mathbf{X}) &= p(\mathbf{z}_{N+1} \mid \mathbf{z}_N)
\end{aligned}
$$

5.1 Evaluating $\gamma(\mathbf{z}_n)$

Proof: $\gamma(\mathbf{z}_n) = \alpha(\mathbf{z}_n)\beta(\mathbf{z}_n)/p(\mathbf{X})$

$$
\begin{aligned}
\gamma(\mathbf{z}_n) = p(\mathbf{z}_n \mid \mathbf{X}) &= \frac{p(\mathbf{X} \mid \mathbf{z}_n)\, p(\mathbf{z}_n)}{p(\mathbf{X})} && \text{(Bayes' rule)} \\
&= \frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n)\, p(\mathbf{z}_n)}{p(\mathbf{X})} && \text{(cond. indep.)} \\
&= \frac{\dfrac{p(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{z}_n)}{\cancel{p(\mathbf{z}_n)}}\, \cancel{p(\mathbf{z}_n)}\, p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n)}{p(\mathbf{X})} && \text{(product rule)} \\
&= \frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{z}_n)\, p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n)}{p(\mathbf{X})} = \frac{\alpha(\mathbf{z}_n)\, \beta(\mathbf{z}_n)}{p(\mathbf{X})}. && \blacksquare
\end{aligned}
$$

(We drop $\boldsymbol{\theta}^{\text{old}}$ from the notation in this section.)

Definition : The Forward and Backward Variables

$$
\alpha(\mathbf{z}_n) \equiv p(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{z}_n), \qquad \beta(\mathbf{z}_n) \equiv p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n).
$$

- $\alpha(\mathbf{z}_n)$ is the **joint probability** of observing all the data up to time n **and** the value of $\mathbf{z}_n$ ("the past").
- $\beta(\mathbf{z}_n)$ is the **conditional probability** of all future data from $n+1$ to N given the value of $\mathbf{z}_n$ ("the future").
- Each is a set of K numbers, one per setting of the 1-of-K vector $\mathbf{z}_n$. Neither sums to one in general.

5.2 The Forward Recursion

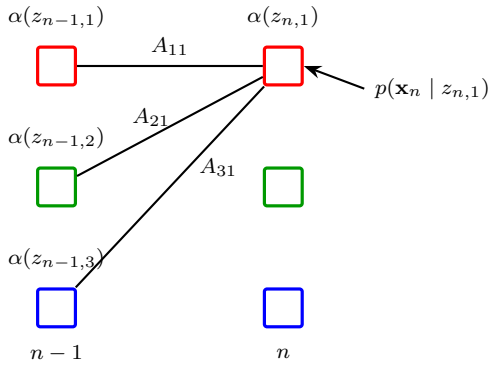
Key Rule: Forward (α) Recursion

$$\boxed{\alpha(\mathbf{z}_n) = p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \alpha(\mathbf{z}_{n-1})\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1})} \quad \text{i.e.} \quad \alpha(z_{nk}) = p(\mathbf{x}_n \mid \boldsymbol{\phi}_k) \sum_{j=1}^{K} \alpha(z_{n-1,j}) A_{jk}.$$

Initialization: $\alpha(z_{1k}) = \pi_k\, p(\mathbf{x}_1 \mid \boldsymbol{\phi}_k)$ for $k = 1, \ldots, K$, i.e.

$$\alpha(\mathbf{z}_1) = p(\mathbf{x}_1, \mathbf{z}_1) = p(\mathbf{z}_1)\, p(\mathbf{x}_1 \mid \mathbf{z}_1) = \prod_{k=1}^{K} \{\pi_k\, p(\mathbf{x}_1 \mid \boldsymbol{\phi}_k)\}^{z_{1k}}.$$

In words, $\alpha(z_{n,1})$ is obtained by (1) taking the elements $\alpha(z_{n-1,j})$ at step $n-1$; (2) **summing** them with weights A_{j1}, the values of $p(\mathbf{z}_n \mid \mathbf{z}_{n-1})$; (3) **multiplying** by the data contribution $p(\mathbf{x}_n \mid z_{n,1})$.



One step of the forward recursion on the trellis. *(Redrawn after Bishop.)*

Proof: The Forward Recursion

$$\begin{aligned}
\alpha(\mathbf{z}_n) &= p(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{z}_n) = p(\mathbf{x}_1, \ldots, \mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{z}_n) && \text{(product rule)} \\
&= p(\mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{z}_n)\, p(\mathbf{z}_n) && \text{(cond. indep.)} \\
&= p(\mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}, \mathbf{z}_n) && \text{(product rule)} \\
&= p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}, \mathbf{z}_{n-1}, \mathbf{z}_n) && \text{(marginalization)} \\
&= p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}, \mathbf{z}_n \mid \mathbf{z}_{n-1})\, p(\mathbf{z}_{n-1}) && \text{(product rule)} \\
&= p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} \mid \mathbf{z}_{n-1})\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1})\, p(\mathbf{z}_{n-1}) && \text{(cond. indep.)} \\
&= p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \underbrace{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1}, \mathbf{z}_{n-1})}_{\alpha(\mathbf{z}_{n-1})}\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1}). && \text{(product rule)} \ \blacksquare
\end{aligned}$$

Key Rule: Cost of the Forward Recursion

There are K terms in the sum over $\mathbf{z}_{n-1}$, and the right-hand side must be evaluated for K values of $\mathbf{z}_n$, so each step costs $O(K^2)$. The **total complexity** for the whole chain is $O(K^2N)$ – **linear** in N, compared with $O(K^N)$ for brute-force summation.

5.3 The Backward Recursion

Key Rule: Backward (β) Recursion

$$\boxed{\beta(\mathbf{z}_n) = \sum_{\mathbf{z}_{n+1}} \beta(\mathbf{z}_{n+1})\, p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1})\, p(\mathbf{z}_{n+1} \mid \mathbf{z}_n)}$$

$$\text{i.e.} \quad \beta(z_{nj}) = \sum_{k=1}^{K} A_{jk}\, p(\mathbf{x}_{n+1} \mid \boldsymbol{\phi}_k)\, \beta(z_{n+1,k}).$$

Initialization: $\beta(\mathbf{z}_N) = 1$ for all settings of $\mathbf{z}_N$.
In words, $\beta(z_{n1})$ is obtained by (1) taking the components $\beta(z_{n+1,k})$ at step $n+1$; (2) **summing** them with weights given by the **products** of A_{1k} (values of $p(\mathbf{z}_{n+1} \mid \mathbf{z}_n)$) and (3) the corresponding values of the emission density $p(\mathbf{x}_{n+1} \mid z_{n+1,k})$. Cost: again $O(K^2)$ per step.

Proof: The Backward Recursion

$$\begin{aligned}
\beta(\mathbf{z}_n) &= p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n) = \sum_{\mathbf{z}_{n+1}} p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N, \mathbf{z}_{n+1} \mid \mathbf{z}_n) && \text{(marginalization)} \\
&= \sum_{\mathbf{z}_{n+1}} p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n, \mathbf{z}_{n+1})\, p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) && \text{(product rule)} \\
&= \sum_{\mathbf{z}_{n+1}} p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_{n+1})\, p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) && \text{(cond. indep.)} \\
&= \sum_{\mathbf{z}_{n+1}} \underbrace{p(\mathbf{x}_{n+2}, \ldots, \mathbf{x}_N \mid \mathbf{z}_{n+1})}_{\beta(\mathbf{z}_{n+1})}\, p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1})\, p(\mathbf{z}_{n+1} \mid \mathbf{z}_n). && \text{(cond. indep.)}\ \blacksquare
\end{aligned}$$

Why $\beta(\mathbf{z}_N) = 1$. Marginalizing $\gamma(\mathbf{z}_n) = \alpha(\mathbf{z}_n)\beta(\mathbf{z}_n)/p(\mathbf{X})$ over $\mathbf{z}_n$ gives

$$p(\mathbf{X}) = \sum_{\mathbf{z}_n} \alpha(\mathbf{z}_n)\, \beta(\mathbf{z}_n) \quad \text{for every } n.$$

At $n = N$, $p(\mathbf{X}) = \sum_{\mathbf{z}_N} p(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{z}_N) = \sum_{\mathbf{z}_N} \alpha(\mathbf{z}_N)$, which is consistent only if $\beta(\mathbf{z}_N) = 1$. □

Scribe's note: equivalently, $\beta(\mathbf{z}_N)$ is the probability of an **empty** set of future observations, which is 1.

5.4 Evaluating $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$

Key Rule: The Pairwise Posterior

$\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$ gives $p(\mathbf{z}_{n-1}, \mathbf{z}_n \mid \mathbf{X})$ for each of the $K \times K$ settings of $(\mathbf{z}_{n-1}, \mathbf{z}_n)$:

$$\xi(\mathbf{z}_{n-1}, \mathbf{z}_n) = p(\mathbf{z}_{n-1}, \mathbf{z}_n \mid \mathbf{X}) = \frac{p(\mathbf{X} \mid \mathbf{z}_{n-1}, \mathbf{z}_n)\, p(\mathbf{z}_{n-1}, \mathbf{z}_n)}{p(\mathbf{X})} \quad \text{(Bayes' rule)}$$

$$= \frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1} | \mathbf{z}_{n-1})\, p(\mathbf{x}_n | \mathbf{z}_n)\, p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N | \mathbf{z}_n)\, p(\mathbf{z}_n | \mathbf{z}_{n-1})\, p(\mathbf{z}_{n-1})}{p(\mathbf{X})} \quad \text{(cond. indep.)}$$

$$= \boxed{\frac{\overbrace{\alpha(\mathbf{z}_{n-1})}^{\text{forward}}\, p(\mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1})\, \overbrace{\beta(\mathbf{z}_n)}^{\text{backward}}}{p(\mathbf{X})}}, \qquad p(\mathbf{X}) = \sum_{\mathbf{z}_N} \alpha(\mathbf{z}_N).$$

In components: $\xi(z_{n-1,j}, z_{nk}) = \alpha(z_{n-1,j})\, A_{jk}\, p(\mathbf{x}_n \mid \boldsymbol{\phi}_k)\, \beta(z_{nk}) / p(\mathbf{X})$. Note the factor A_{jk}, which is why a zero initial A_{jk} stays zero.

Algorithm: Baum-Welch (EM with Forward-Backward) for an HMM

1: **Initialize** $\boldsymbol{\theta}^{\text{old}} = \{\boldsymbol{\pi}, \mathbf{A}, \boldsymbol{\phi}\}$ randomly, respecting sum-to-one and **nonzero** entries
2: **repeat**
3: **Forward:** $\alpha(\mathbf{z}_1) = \boldsymbol{\pi} \odot p(\mathbf{x}_1 \mid \boldsymbol{\phi})$; for $n = 2..N$: $\alpha(\mathbf{z}_n) = p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \alpha(\mathbf{z}_{n-1}) p(\mathbf{z}_n \mid \mathbf{z}_{n-1})$
4: **Backward:** $\beta(\mathbf{z}_N) = 1$; for $n = N-1..1$: $\beta(\mathbf{z}_n) = \sum_{\mathbf{z}_{n+1}} \beta(\mathbf{z}_{n+1}) p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1}) p(\mathbf{z}_{n+1} \mid \mathbf{z}_n)$
5: **E-step:** $p(\mathbf{X}) = \sum_{\mathbf{z}_N} \alpha(\mathbf{z}_N)$, $\gamma(\mathbf{z}_n) = \alpha(\mathbf{z}_n)\beta(\mathbf{z}_n)/p(\mathbf{X})$, $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n) = \alpha(\mathbf{z}_{n-1}) p(\mathbf{x}_n \mid \mathbf{z}_n) p(\mathbf{z}_n \mid \mathbf{z}_{n-1}) \beta(\mathbf{z}_n) / p(\mathbf{X})$
6: **M-step:** $\pi_k \propto \gamma(z_{1k})$, $A_{jk} \propto \sum_{n=2}^{N} \xi(z_{n-1,j}, z_{nk})$, $\boldsymbol{\phi}_k$ from the γ-weighted statistics (Section 4.2)
7: **until** $\ln p(\mathbf{X} \mid \boldsymbol{\theta})$ (or $\boldsymbol{\theta}$) has converged

Each iteration costs $O(K^2 N)$; as in Lecture 7, $p(\mathbf{X} \mid \boldsymbol{\theta})$ never decreases. In practice the scaled recursions of Section 7 are used.

Worked Example: Forward-Backward on the Two-Coin HMM (Scribe's Example)

$\mathbf{X} = (\mathrm{H}, \mathrm{T}, \mathrm{H})$, with $\boldsymbol{\pi}$, $\mathbf{A}$ and emissions from Section 3. Vectors are written as (state 1, state 2).

Forward pass.

$$\alpha(\mathbf{z}_1) = (0.5 \cdot 0.8,\ 0.5 \cdot 0.2) = (0.4,\ 0.1)$$

$$\alpha(\mathbf{z}_2) = \big(0.2\,(0.4 \cdot 0.7 + 0.1 \cdot 0.3),\ \ 0.8\,(0.4 \cdot 0.3 + 0.1 \cdot 0.7)\big)$$
$$= (0.2 \cdot 0.31,\ 0.8 \cdot 0.19) = (0.062,\ 0.152)$$

$$\alpha(\mathbf{z}_3) = \big(0.8\,(0.062 \cdot 0.7 + 0.152 \cdot 0.3),\ \ 0.2\,(0.062 \cdot 0.3 + 0.152 \cdot 0.7)\big)$$
$$= (0.8 \cdot 0.089,\ 0.2 \cdot 0.125) = (0.0712,\ 0.0250)$$

Likelihood: $p(\mathbf{X}) = 0.0712 + 0.0250 = 0.0962$ – the same as summing all 8 path probabilities

by brute force.

Backward pass.

$$\beta(\mathbf{z}_3) = (1,\ 1)$$
$$\beta(\mathbf{z}_2) = (0.7 \cdot 0.8 + 0.3 \cdot 0.2,\ \ 0.3 \cdot 0.8 + 0.7 \cdot 0.2) = (0.62,\ 0.38)$$
$$\beta(\mathbf{z}_1) = (0.7 \cdot 0.2 \cdot 0.62 + 0.3 \cdot 0.8 \cdot 0.38,\ \ 0.3 \cdot 0.2 \cdot 0.62 + 0.7 \cdot 0.8 \cdot 0.38) = (0.178,\ 0.250)$$

Check: $\sum_{\mathbf{z}_1} \alpha(\mathbf{z}_1)\beta(\mathbf{z}_1) = 0.4 \cdot 0.178 + 0.1 \cdot 0.25 = 0.0962 = p(\mathbf{X})$ ✓

Posteriors $\gamma(\mathbf{z}_n) = \alpha(\mathbf{z}_n)\beta(\mathbf{z}_n)/p(\mathbf{X})$:

	$n=1$ (H)	$n=2$ (T)	$n=3$ (H)
$\gamma(z_{n1})$	$0.0712/0.0962 = 0.740$	$0.03844/0.0962 = 0.400$	0.740
$\gamma(z_{n2})$	0.260	0.600	0.260
most probable state	1	**2**	1

Pairwise posteriors $\xi(z_{n-1,j}, z_{nk}) = \alpha(z_{n-1,j})A_{jk}p(\mathbf{x}_n \mid k)\beta(z_{nk})/p(\mathbf{X})$ (rows j = state at $n-1$, columns k = state at n):

$$\xi(\mathbf{z}_1, \mathbf{z}_2) = \frac{1}{0.0962}\begin{bmatrix} 0.4 \cdot 0.7 \cdot 0.2 \cdot 0.62 & 0.4 \cdot 0.3 \cdot 0.8 \cdot 0.38 \\ 0.1 \cdot 0.3 \cdot 0.2 \cdot 0.62 & 0.1 \cdot 0.7 \cdot 0.8 \cdot 0.38 \end{bmatrix} = \begin{bmatrix} 0.361 & 0.379 \\ 0.039 & 0.221 \end{bmatrix},$$

$$\xi(\mathbf{z}_2, \mathbf{z}_3) = \frac{1}{0.0962}\begin{bmatrix} 0.062 \cdot 0.7 \cdot 0.8 \cdot 1 & 0.062 \cdot 0.3 \cdot 0.2 \cdot 1 \\ 0.152 \cdot 0.3 \cdot 0.8 \cdot 1 & 0.152 \cdot 0.7 \cdot 0.2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 0.361 & 0.039 \\ 0.379 & 0.221 \end{bmatrix}.$$

Each matrix sums to 1; its row sums are $\gamma(\mathbf{z}_{n-1})$ and its column sums are $\gamma(\mathbf{z}_n)$ (e.g. $0.361 + 0.379 = 0.740$).

Worked Example: One Baum-Welch M-Step on the Two-Coin HMM (Scribe's Example)

Using the γ and ξ above:

$$\boldsymbol{\pi}^{\text{new}} = \gamma(\mathbf{z}_1) = (0.740,\ 0.260),$$
$$A_{11}^{\text{new}} = \frac{\xi_2(1,1) + \xi_3(1,1)}{\sum_k [\xi_2(1,k) + \xi_3(1,k)]} = \frac{0.361 + 0.361}{0.740 + 0.400} = 0.633, \qquad A_{12}^{\text{new}} = 0.367,$$
$$A_{21}^{\text{new}} = \frac{0.039 + 0.379}{0.260 + 0.600} = 0.486, \qquad A_{22}^{\text{new}} = 0.514,$$
$$p^{\text{new}}(\text{H} \mid k{=}1) = \frac{\gamma(z_{11}) + \gamma(z_{31})}{\gamma(z_{11}) + \gamma(z_{21}) + \gamma(z_{31})} = \frac{0.740 + 0.740}{0.740 + 0.400 + 0.740} = 0.787,$$
$$p^{\text{new}}(\text{H} \mid k{=}2) = \frac{0.260 + 0.260}{0.260 + 0.600 + 0.260} = 0.464.$$

Re-running the forward pass with $\boldsymbol{\theta}^{\text{new}}$ gives $p(\mathbf{X} \mid \boldsymbol{\theta}^{\text{new}}) = 0.153 > 0.0962$: the likelihood increased, as EM guarantees. (With only three tosses EM will happily overfit; this is just to make the updates concrete.)

5.5 The Predictive Distribution

Key Rule: Predicting $\mathbf{x}_{N+1}$

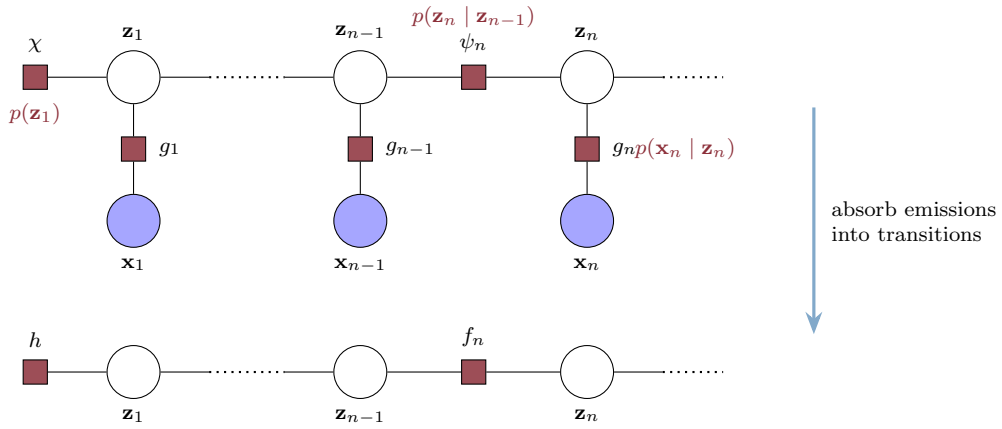
Given $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, predict $\mathbf{x}_{N+1}$ (e.g. financial forecasting):

$$\begin{aligned} p(\mathbf{x}_{N+1} \mid \mathbf{X}) &= \sum_{\mathbf{z}_{N+1}} p(\mathbf{x}_{N+1}, \mathbf{z}_{N+1} \mid \mathbf{X}) = \sum_{\mathbf{z}_{N+1}} p(\mathbf{x}_{N+1} \mid \mathbf{z}_{N+1})\, p(\mathbf{z}_{N+1} \mid \mathbf{X}) \\ &= \sum_{\mathbf{z}_{N+1}} p(\mathbf{x}_{N+1} \mid \mathbf{z}_{N+1}) \sum_{\mathbf{z}_N} p(\mathbf{z}_{N+1} \mid \mathbf{z}_N)\, p(\mathbf{z}_N \mid \mathbf{X}) \\ &= \frac{1}{p(\mathbf{X})} \sum_{\mathbf{z}_{N+1}} p(\mathbf{x}_{N+1} \mid \mathbf{z}_{N+1}) \sum_{\mathbf{z}_N} p(\mathbf{z}_{N+1} \mid \mathbf{z}_N)\, \alpha(\mathbf{z}_N), \end{aligned}$$

(marginalization, product rule and the last two CI properties of Section 5, then $p(\mathbf{z}_N \mid \mathbf{X}) = \alpha(\mathbf{z}_N)/p(\mathbf{X})$.) Only the **forward** recursion is needed: propagate $\alpha(\mathbf{z}_N)/p(\mathbf{X}) = p(\mathbf{z}_N \mid \mathbf{X})$ one step through $\mathbf{A}$, then through the emissions.
Scribe's check on the running example: $p(\mathbf{z}_3 \mid \mathbf{X}) = (0.740, 0.260)$; one step through $\mathbf{A}$ gives $p(\mathbf{z}_4 \mid \mathbf{X}) = (0.596, 0.404)$, so $p(\mathbf{x}_4 = \mathrm{H} \mid \mathbf{X}) = 0.8 \cdot 0.596 + 0.2 \cdot 0.404 = 0.558$.

6 Forward-Backward as the Sum-Product Algorithm

The forward-backward recursions are not new: they are the **sum-product** algorithm of Lectures 4–5 run on the HMM's **factor graph**.



Top: factor graph of the HMM (prior χ, transitions ψ_n, emissions g_n). Bottom: simplified chain. *(Redrawn after Bishop.)*

Definition : Simplified HMM Factor Graph

Since we are always **conditioning** on the observed $\mathbf{x}_1, \ldots, \mathbf{x}_N$, the emission factors are just functions of $\mathbf{z}_n$ and can be **absorbed** into the transition factors:

$$h(\mathbf{z}_1) = p(\mathbf{z}_1)\, p(\mathbf{x}_1 \mid \mathbf{z}_1), \qquad f_n(\mathbf{z}_{n-1}, \mathbf{z}_n) = p(\mathbf{z}_n \mid \mathbf{z}_{n-1})\, p(\mathbf{x}_n \mid \mathbf{z}_n).$$

The result is a simple chain $h - \mathbf{z}_1 - f_2 - \mathbf{z}_2 - \cdots - f_N - \mathbf{z}_N$.

Key Rule: Messages towards an Arbitrary Root $\mathbf{z}_n$ Recover α and β

From the left. The node-to-factor message needs **no computation** since each variable node has only two neighbours: $\mu_{\mathbf{z}_{n-1} \to f_n}(\mathbf{z}_{n-1}) = \mu_{f_{n-1} \to \mathbf{z}_{n-1}}(\mathbf{z}_{n-1})$. The factor-to-node message is then

$$\underbrace{\mu_{f_n \to \mathbf{z}_n}(\mathbf{z}_n)}_{\alpha(\mathbf{z}_n)} = \sum_{\mathbf{z}_{n-1}} \underbrace{f_n(\mathbf{z}_{n-1}, \mathbf{z}_n)}_{p(\mathbf{z}_n \mid \mathbf{z}_{n-1}) p(\mathbf{x}_n \mid \mathbf{z}_n)} \underbrace{\mu_{f_{n-1} \to \mathbf{z}_{n-1}}(\mathbf{z}_{n-1})}_{\alpha(\mathbf{z}_{n-1})}$$

$$\implies \quad \alpha(\mathbf{z}_n) = p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \alpha(\mathbf{z}_{n-1})\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1}),$$

the same as the forward recursion! (Starting message: $\mu_{h \to \mathbf{z}_1} = h(\mathbf{z}_1) = \alpha(\mathbf{z}_1)$.)
From the right. Likewise $\mu_{\mathbf{z}_{n+1} \to f_{n+1}}(\mathbf{z}_{n+1}) = \mu_{f_{n+2} \to \mathbf{z}_{n+1}}(\mathbf{z}_{n+1})$, and

$$\underbrace{\mu_{f_{n+1} \to \mathbf{z}_n}(\mathbf{z}_n)}_{\beta(\mathbf{z}_n)} = \sum_{\mathbf{z}_{n+1}} \underbrace{f_{n+1}(\mathbf{z}_n, \mathbf{z}_{n+1})}_{p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1})} \underbrace{\mu_{f_{n+2} \to \mathbf{z}_{n+1}}(\mathbf{z}_{n+1})}_{\beta(\mathbf{z}_{n+1})}$$

$$\implies \quad \beta(\mathbf{z}_n) = \sum_{\mathbf{z}_{n+1}} \beta(\mathbf{z}_{n+1})\, p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1})\, p(\mathbf{z}_{n+1} \mid \mathbf{z}_n),$$

the same as the backward recursion! (The rightmost leaf $\mathbf{z}_N$ sends the unit message, i.e. $\beta(\mathbf{z}_N) = 1$.)

At the root, the product of incoming messages is $\alpha(\mathbf{z}_n)\beta(\mathbf{z}_n) = p(\mathbf{z}_n, \mathbf{X})$; normalizing gives $\gamma(\mathbf{z}_n)$, as sum-product always gives the (unnormalized) marginal at the root.

7 Scaling Factors

Key Difficulty: Numerical Underflow

Each step obtains $\alpha(\mathbf{z}_n)$ from $\alpha(\mathbf{z}_{n-1})$ by multiplying by $p(\mathbf{z}_n \mid \mathbf{z}_{n-1})$ and $p(\mathbf{x}_n \mid \mathbf{z}_n)$. These are often **significantly less than unity**, so as we work forward along the chain $\alpha(\mathbf{z}_n)$ goes to **zero exponentially quickly** (already $p(\mathbf{X}) = 0.0962$ after only three tosses in our example; for N in the hundreds it underflows double precision).
Taking logarithms does not help directly, because we are forming **sums** of products of small numbers:

$$\log \alpha(\mathbf{z}_n) = \log p(\mathbf{x}_n \mid \mathbf{z}_n) + \log \underbrace{\sum_{\mathbf{z}_{n-1}} \alpha(\mathbf{z}_{n-1})\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1})}_{\text{small number}},$$

and the log of a sum does not split. We therefore work with **re-scaled** versions of $\alpha(\mathbf{z}_n)$ whose values remain of order unity.
Scribe's note: one *can* stay in the log domain using the log-sum-exp trick; the scaling approach below is the standard one because it also hands us $p(\mathbf{z}_n \mid \mathbf{x}_1, \ldots, \mathbf{x}_n)$ and $\ln p(\mathbf{X})$ for free.

Definition : Normalized Forward Variable and Scaling Factor

$$\widehat{\alpha}(\mathbf{z}_n) = p(\mathbf{z}_n \mid \mathbf{x}_1, \ldots, \mathbf{x}_n) = \frac{\alpha(\mathbf{z}_n)}{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}, \qquad c_n = p(\mathbf{x}_n \mid \mathbf{x}_1, \ldots, \mathbf{x}_{n-1}).$$

$\widehat{\alpha}(\mathbf{z}_n)$ is **well behaved numerically** because it is a probability distribution over K values for any n (it is the **filtering** distribution).

Key Rule: Scaled Forward Recursion

Divide the forward recursion by $p(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ and write $p(\mathbf{x}_1, \ldots, \mathbf{x}_n) = c_n\, p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})$:

$$\underbrace{\frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})}}_{c_n} \underbrace{\frac{\alpha(\mathbf{z}_n)}{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}}_{\widehat{\alpha}(\mathbf{z}_n)} = p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \underbrace{\frac{\alpha(\mathbf{z}_{n-1})}{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})}}_{\widehat{\alpha}(\mathbf{z}_{n-1})} p(\mathbf{z}_n \mid \mathbf{z}_{n-1}),$$

$$\boxed{c_n\, \widehat{\alpha}(\mathbf{z}_n) = p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \widehat{\alpha}(\mathbf{z}_{n-1})\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1}) \equiv \widetilde{\alpha}(\mathbf{z}_n), \qquad c_n = \sum_{\mathbf{z}_n} \widetilde{\alpha}(\mathbf{z}_n)}$$

i.e. compute the unnormalized $\widetilde{\alpha}(\mathbf{z}_n)$, then c_n is the sum of its K entries and $\widehat{\alpha}(\mathbf{z}_n) = \widetilde{\alpha}(\mathbf{z}_n)/c_n$.

Proof: $c_n = \sum_{\mathbf{z}_n} \widetilde{\alpha}(\mathbf{z}_n)$

$$\begin{aligned}\sum_{\mathbf{z}_n} \widetilde{\alpha}(\mathbf{z}_n) &= \sum_{\mathbf{z}_n} p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \widehat{\alpha}(\mathbf{z}_{n-1})\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1}) \\ &= \sum_{\mathbf{z}_n} p(\mathbf{x}_n \mid \mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \frac{\alpha(\mathbf{z}_{n-1})}{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})}\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1}) \\ &= \sum_{\mathbf{z}_n} \frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_n, \mathbf{z}_n)}{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})} = \frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})} = c_n,\end{aligned}$$

using the forward recursion $p(\mathbf{x}_n|\mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \alpha(\mathbf{z}_{n-1}) p(\mathbf{z}_n|\mathbf{z}_{n-1}) = \alpha(\mathbf{z}_n)$ in the second line. □

Key Rule: Scaled Backward Recursion

Define the normalized backward variable

$$\widehat{\beta}(\mathbf{z}_n) = \frac{\beta(\mathbf{z}_n)}{p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{x}_1, \ldots, \mathbf{x}_n)} = \frac{p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{z}_n)}{p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{x}_1, \ldots, \mathbf{x}_n)}.$$

The backward recursion in normalized form is

$$\boxed{c_{n+1}\, \widehat{\beta}(\mathbf{z}_n) = \sum_{\mathbf{z}_{n+1}} \widehat{\beta}(\mathbf{z}_{n+1})\, p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1})\, p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) \equiv \widetilde{\beta}(\mathbf{z}_n)}$$

where c_{n+1} is **stored and reused from the forward recursion**, since

$$c_{n+1} = \frac{p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{x}_1, \ldots, \mathbf{x}_n)}{p(\mathbf{x}_{n+2}, \ldots, \mathbf{x}_N \mid \mathbf{x}_1, \ldots, \mathbf{x}_{n+1})} = \frac{p(\mathbf{X})/p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}{p(\mathbf{X})/p(\mathbf{x}_1, \ldots, \mathbf{x}_{n+1})} = \frac{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n+1})}{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}.$$

Unlike $\widehat{\alpha}$, $\widehat{\beta}(\mathbf{z}_n)$ is a **ratio** of probabilities, not a distribution over $\mathbf{z}_n$, but it remains of order unity.

Key Rule: Posteriors from the Scaled Variables

$$\boxed{\gamma(\mathbf{z}_n) = \widehat{\alpha}(\mathbf{z}_n)\, \widehat{\beta}(\mathbf{z}_n), \qquad \xi(\mathbf{z}_{n-1}, \mathbf{z}_n) = c_n^{-1}\, \widehat{\alpha}(\mathbf{z}_{n-1})\, p(\mathbf{x}_n \mid \mathbf{z}_n)\, p(\mathbf{z}_n \mid \mathbf{z}_{n-1})\, \widehat{\beta}(\mathbf{z}_n)}$$

Proof for γ. Using $p(\mathbf{x}_{n+1}, \ldots, \mathbf{x}_N \mid \mathbf{x}_1, \ldots, \mathbf{x}_n) = p(\mathbf{x}_1, \ldots, \mathbf{x}_N)/p(\mathbf{x}_1, \ldots, \mathbf{x}_n)$,

$$\widehat{\alpha}(\mathbf{z}_n)\widehat{\beta}(\mathbf{z}_n) = \frac{\alpha(\mathbf{z}_n)}{\cancel{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}} \cdot \frac{\beta(\mathbf{z}_n)}{p(\mathbf{x}_1, \ldots, \mathbf{x}_N)/\cancel{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}} = \frac{\alpha(\mathbf{z}_n)\beta(\mathbf{z}_n)}{p(\mathbf{X})}. \qquad \square$$

Proof for ξ. With $c_n^{-1} = p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})/p(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ and $\widehat{\beta}(\mathbf{z}_n) = \beta(\mathbf{z}_n)\, p(\mathbf{x}_1, \ldots, \mathbf{x}_n)/p(\mathbf{X})$,

$$\begin{aligned}&c_n^{-1}\, \widehat{\alpha}(\mathbf{z}_{n-1})\, p(\mathbf{x}_n|\mathbf{z}_n)\, p(\mathbf{z}_n|\mathbf{z}_{n-1})\, \widehat{\beta}(\mathbf{z}_n) \\ &= \frac{\cancel{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})}}{\cancel{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}} \cdot \frac{\alpha(\mathbf{z}_{n-1})}{\cancel{p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})}}\, p(\mathbf{x}_n|\mathbf{z}_n)\, p(\mathbf{z}_n|\mathbf{z}_{n-1}) \cdot \frac{\beta(\mathbf{z}_n)\, \cancel{p(\mathbf{x}_1, \ldots, \mathbf{x}_n)}}{p(\mathbf{X})} \\ &= \frac{\alpha(\mathbf{z}_{n-1})\, p(\mathbf{x}_n|\mathbf{z}_n)\, p(\mathbf{z}_n|\mathbf{z}_{n-1})\, \beta(\mathbf{z}_n)}{p(\mathbf{X})}. \qquad \square\end{aligned}$$

Key Rule: Initialization of the Scaled Recursions

- $\widehat{\alpha}(\mathbf{z}_1) = \dfrac{\alpha(\mathbf{z}_1)}{p(\mathbf{x}_1)} = \dfrac{p(\mathbf{x}_1, \mathbf{z}_1)}{\sum_{\mathbf{z}_1} p(\mathbf{x}_1, \mathbf{z}_1)}$: divide $\alpha(\mathbf{z}_1)$ by the **sum of all its values** (so $c_1 = \sum_{\mathbf{z}_1} \alpha(\mathbf{z}_1) = p(\mathbf{x}_1)$).
- $\widehat{\beta}(\mathbf{z}_N) = 1$. **Proof:** $\gamma(\mathbf{z}_N) = \widehat{\alpha}(\mathbf{z}_N)\widehat{\beta}(\mathbf{z}_N)$, and $\widehat{\beta}(\mathbf{z}_N) = \beta(\mathbf{z}_N)/\eta$ with $\beta(\mathbf{z}_N) = 1$ and some normalizer η (written n on the slide – not the time index). Marginalizing over $\mathbf{z}_N$: $\underbrace{\sum_{\mathbf{z}_N} \gamma(\mathbf{z}_N)}_{1} = \sum_{\mathbf{z}_N} \widehat{\alpha}(\mathbf{z}_N) \cdot \frac{1}{\eta}$, so $\eta = \sum_{\mathbf{z}_N} \widehat{\alpha}(\mathbf{z}_N) = 1$ and $\widehat{\beta}(\mathbf{z}_N) = 1$. □

Key Rule: The Log-Likelihood from the Scaling Factors (Scribe's Addition)

Telescoping the definition $c_n = p(\mathbf{x}_1, \ldots, \mathbf{x}_n)/p(\mathbf{x}_1, \ldots, \mathbf{x}_{n-1})$ gives

$$p(\mathbf{X}) = \prod_{n=1}^{N} c_n \qquad \Longrightarrow \qquad \ln p(\mathbf{X}) = \sum_{n=1}^{N} \ln c_n,$$

which is how the Baum-Welch convergence test is evaluated in practice without ever forming the tiny number $p(\mathbf{X})$ (Bishop eq. 13.63).

Worked Example: Scaled Recursions on the Two-Coin HMM (Scribe's Example)

Forward. $\widetilde{\alpha}(\mathbf{z}_1) = (0.4, 0.1)$, $c_1 = 0.5$, $\widehat{\alpha}(\mathbf{z}_1) = (0.8, 0.2)$.
$\widetilde{\alpha}(\mathbf{z}_2) = \big(0.2(0.8 \cdot 0.7 + 0.2 \cdot 0.3),\ 0.8(0.8 \cdot 0.3 + 0.2 \cdot 0.7)\big) = (0.124, 0.304)$,
$c_2 = 0.428$, $\widehat{\alpha}(\mathbf{z}_2) = (0.290, 0.710)$.
$\widetilde{\alpha}(\mathbf{z}_3) = (0.333, 0.117)$, $c_3 = 0.4495$, $\widehat{\alpha}(\mathbf{z}_3) = (0.740, 0.260)$.
Check: $c_1 c_2 c_3 = 0.5 \cdot 0.428 \cdot 0.4495 = 0.0962 = p(\mathbf{X})$ ✓. Every $\widehat{\alpha}$ sums to one, and $\widehat{\alpha}(\mathbf{z}_n)$ equals $\alpha(\mathbf{z}_n)$ divided by its sum, e.g. $(0.062, 0.152)/0.214 = (0.290, 0.710)$.
Backward. $\widehat{\beta}(\mathbf{z}_3) = (1, 1)$; $\widehat{\beta}(\mathbf{z}_2) = (0.62, 0.38)/c_3 = (1.379, 0.845)$; $\widehat{\beta}(\mathbf{z}_1) = \widetilde{\beta}(\mathbf{z}_1)/c_2 = (0.925, 1.299)$.
Posteriors. $\gamma(\mathbf{z}_2) = \widehat{\alpha}(\mathbf{z}_2) \odot \widehat{\beta}(\mathbf{z}_2) = (0.290 \cdot 1.379,\ 0.710 \cdot 0.845) = (0.400, 0.600)$ – identical to the unscaled result. ✓

8 The Viterbi Algorithm

We now want the **maximal probability** and the **most probable sequence of hidden states** for a given observation sequence.

Key Difficulty: Most Probable Sequence ≠ Sequence of Most Probable States

Recall (Lecture 5): finding the **most probable sequence** of latent states is **not** the same as finding the set of states that are **individually** the most probable (the arg max of each $\gamma(\mathbf{z}_n)$). The latter can even be a sequence of **zero** probability, e.g. if it uses a transition with $A_{jk} = 0$. Finding the most probable *sequence* is solved efficiently by the **max-sum** algorithm, which for HMMs is called the **Viterbi algorithm**.

Key Rule: Max-Sum Messages on the HMM Chain

Choose $\mathbf{z}_N$ as the **root**. With $f_{n+1}(\mathbf{z}_n, \mathbf{z}_{n+1}) = p(\mathbf{z}_{n+1} \mid \mathbf{z}_n)\, p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1})$, the max-sum messages are

$$\begin{aligned} &\text{Node-to-factor:} && \mu_{\mathbf{z}_n \to f_{n+1}}(\mathbf{z}_n) = \mu_{f_n \to \mathbf{z}_n}(\mathbf{z}_n), \\ &\text{Factor-to-node:} && \mu_{f_{n+1} \to \mathbf{z}_{n+1}}(\mathbf{z}_{n+1}) = \max_{\mathbf{z}_n} \left\{ \ln f_{n+1}(\mathbf{z}_n, \mathbf{z}_{n+1}) + \mu_{\mathbf{z}_n \to f_{n+1}}(\mathbf{z}_n) \right\}. \end{aligned}$$

- This is the forward recursion with the **summation over $\mathbf{z}_n$ replaced by a max** (and products by sums of logs).
- **No backward pass** is needed for the max probability, since it is **the same regardless of the choice of root node**.

Definition : The Viterbi Recursion

Denote $\omega(\mathbf{z}_n) \equiv \mu_{f_n \to \mathbf{z}_n}(\mathbf{z}_n)$ ($K \times 1$ entries). Then

$$\boxed{\omega(\mathbf{z}_{n+1}) = \ln p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1}) + \max_{\mathbf{z}_n} \left\{ \ln p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) + \omega(\mathbf{z}_n) \right\}}$$

which can be **computed recursively**, with **initialization** given by the factor-to-node message from h:

$$\omega(\mathbf{z}_1) = \ln p(\mathbf{z}_1) + \ln p(\mathbf{x}_1 \mid \mathbf{z}_1).$$

At the **root**, the maximal probability of the joint distribution is

$$\max_{\mathbf{z}_1, \ldots, \mathbf{z}_N} \ln p(\mathbf{x}_1, \ldots, \mathbf{x}_N, \mathbf{z}_1, \ldots, \mathbf{z}_N) = \max_{\mathbf{z}_N} \omega(\mathbf{z}_N).$$

(The slide writes $\max p(\cdot) = \max \omega(\mathbf{z}_N)$; since ω lives in the log domain, strictly $\max p = \exp\{\max \omega\}$.)

Proof: The Viterbi Recursion

Substituting f_{n+1} and the node-to-factor message $\mu_{\mathbf{z}_n \to f_{n+1}}(\mathbf{z}_n) = \mu_{f_n \to \mathbf{z}_n}(\mathbf{z}_n) = \omega(\mathbf{z}_n)$,

$$\begin{aligned}\omega(\mathbf{z}_{n+1}) = \mu_{f_{n+1} \to \mathbf{z}_{n+1}}(\mathbf{z}_{n+1}) &= \max_{\mathbf{z}_n} \big\{ \ln p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) + \ln p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1}) + \omega(\mathbf{z}_n) \big\} \\ &= \ln p(\mathbf{x}_{n+1} \mid \mathbf{z}_{n+1}) + \max_{\mathbf{z}_n} \big\{ \ln p(\mathbf{z}_{n+1} \mid \mathbf{z}_n) + \omega(\mathbf{z}_n) \big\},\end{aligned}$$

pulling out the emission term because it does not depend on $\mathbf{z}_n$ (the $(\max, +)$ distributive law of Lecture 5). ■

Key Rule: No Scaling Needed

The Viterbi algorithm works with **log probabilities** and only **max** and **+** – no log of a sum appears – so there is **no need for scaling**: $\omega(\mathbf{z}_n)$ just becomes a large negative number, which is harmless.

Key Rule: Back-Tracking for the Most Probable Sequence

The forward pass to $\mathbf{z}_N$ gives the maximal probability of $p(\mathbf{X}, \mathbf{Z})$; we also want the **sequence of latent values** that achieves it. Use the **back-tracking** procedure of Lecture 5:

- **Keep a record** of the value of $\mathbf{z}_n$ that achieves the maximum for each of the K values of $\mathbf{z}_{n+1}$:

$$\psi(k_{n+1}) = \arg\max_{\mathbf{z}_n} \big\{ \ln p(\mathbf{z}_{n+1}{=}k \mid \mathbf{z}_n) + \omega(\mathbf{z}_n) \big\}, \qquad k = 1, \ldots, K,$$

  stored as a $K \times N$ table.
- At the end of the chain (the root $\mathbf{z}_N$), take $k_N^{\max} = \arg\max_k \omega(z_{Nk})$, then **backtrack recursively**:

$$\boxed{k_n^{\max} = \psi(k_{n+1}^{\max})}, \qquad n = N-1, \ldots, 1.$$

Each back-pointer is followed from an already-fixed successor, so the result is a **consistent** path, even when ties exist.

Notation warning: ψ is overloaded – in Section 6 ψ_n is the transition factor, and Lecture 5 called the back-pointer table δ_{ji}. On the slide's trellis, a state labelled $\psi(\cdot) = \emptyset$ is one whose back-pointer is not drawn because it lies on no surviving path; strictly, every state at every step has a best predecessor.

Algorithm: Viterbi for an HMM

$\omega(z_{1k}) \leftarrow \ln \pi_k + \ln p(\mathbf{x}_1 \mid \boldsymbol{\phi}_k)$ for $k = 1..K$
for $n = 1$ **to** $N-1$ **do**
  for $k = 1$ **to** K **do**
    $\omega(z_{n+1,k}) \leftarrow \ln p(\mathbf{x}_{n+1} \mid \boldsymbol{\phi}_k) + \max_j \{\ln A_{jk} + \omega(z_{nj})\}$
    $\psi(k_{n+1}) \leftarrow \arg\max_j \{\ln A_{jk} + \omega(z_{nj})\}$ ▷ back-pointer
  end for
end for
$\ln p^{\max} \leftarrow \max_k \omega(z_{Nk})$, $k_N^{\max} \leftarrow \arg\max_k \omega(z_{Nk})$
for $n = N-1$ **down to** 1 **do**

10: $k_n^{\max} \leftarrow \psi(k_{n+1}^{\max})$
11: **end for**
Cost $O(K^2N)$ time and $O(KN)$ memory for the back-pointers – the same as forward-backward.

Worked Example: Viterbi on the Two-Coin HMM (Scribe's Example)

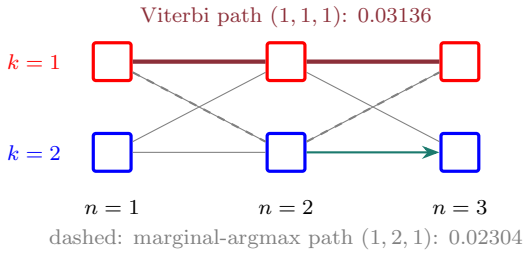
For readability we track e^{ω} (products instead of sums of logs); the arg maxes are identical.

n	k	$e^{\omega(z_{nk})}$	$\psi(k_n)$
1 (H)	1	$0.5 \cdot 0.8 = 0.4$	–
	2	$0.5 \cdot 0.2 = 0.1$	–
2 (T)	1	$0.2 \max(0.4 \cdot 0.7,\ 0.1 \cdot 0.3) = 0.2 \cdot 0.28 = 0.056$	1
	2	$0.8 \max(0.4 \cdot 0.3,\ 0.1 \cdot 0.7) = 0.8 \cdot 0.12 = 0.096$	1
3 (H)	1	$0.8 \max(0.056 \cdot 0.7,\ 0.096 \cdot 0.3) = 0.8 \cdot 0.0392 = \mathbf{0.03136}$	1
	2	$0.2 \max(0.056 \cdot 0.3,\ 0.096 \cdot 0.7) = 0.2 \cdot 0.0672 = 0.01344$	2

Root: $\max_{\mathbf{z}_3} e^{\omega(\mathbf{z}_3)} = 0.03136$ at $k_3^{\max} = 1$. **Backtrack:** $k_2^{\max} = \psi(1_3) = 1$, $k_1^{\max} = \psi(1_2) = 1$. The Viterbi path is $(1, 1, 1)$ with $p(\mathbf{X}, \mathbf{Z}^*) = 0.03136$, i.e. $p(\mathbf{Z}^* \mid \mathbf{X}) = 0.03136/0.0962 = 0.326$.

Compare with the marginals: the individually most probable states from γ (Section 5) are $(1, \mathbf{2}, 1)$, whose joint probability is only $0.5 \cdot 0.8 \cdot 0.3 \cdot 0.8 \cdot 0.3 \cdot 0.8 = 0.02304$. Coin 2 is the better explanation of the middle T *on average*, but any single path that visits it pays for **two** switches (0.3×0.3), while staying with coin 1 pays only for one unlikely toss (0.2). Brute-force enumeration of all 8 paths confirms $(1, 1, 1)$ is the maximum.

Why back-tracking matters: at $n = 2$, state 2 has the larger ω ($0.096 > 0.056$), yet the optimal path passes through state 1 at $n = 2$. Choosing the best ω at each step would also give the wrong answer – only following ψ back from the root is correct.



Trellis for the two-coin example. Red: the optimal path recovered by back-tracking.

Summary: Forward-Backward vs. Viterbi

	Forward-backward	**Viterbi**
Message-passing view	sum-product on the chain	max-sum on the chain
Semiring	$(\sum, \times)$	$(\max, +)$ in logs
Passes	forward α + backward β	forward ω + back-tracking ψ
Output	$p(\mathbf{X})$, marginals $\gamma(\mathbf{z}_n)$, $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n)$	$\max_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z})$ and $\mathbf{Z}^* = \arg\max_{\mathbf{Z}} p(\mathbf{Z} \mid \mathbf{X})$
Used for	E-step of EM (Baum-Welch), prediction	decoding (e.g. the word sequence in speech)
Numerics	needs scaling (c_n)	no scaling (logs)
Cost	$O(K^2 N)$	$O(K^2 N)$

9 Summary

- Described the joint distribution of a HMM with the transition and emission probabilities.
- Used the EM algorithm for maximum likelihood estimation of the latent variables and unknown parameters in the HMM.
- Used the forward-backward algorithm to evaluate the EM algorithm.
- Applied the Viterbi algorithm to find the maximal probability and its configuration.

Key Takeaways

1. An HMM is a **dynamic mixture model**: discrete latents $\mathbf{z}_n$ form a first-order Markov chain, each emitting $\mathbf{x}_n$. The joint is $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = p(\mathbf{z}_1 \mid \boldsymbol{\pi}) \prod_{n \geq 2} p(\mathbf{z}_n \mid \mathbf{z}_{n-1}, \mathbf{A}) \prod_n p(\mathbf{x}_n \mid \mathbf{z}_n, \boldsymbol{\phi})$, with $A_{jk} = p(z_{nk}{=}1 \mid z_{n-1,j}{=}1)$ (rows sum to one). Equal rows of $\mathbf{A}$ recover the i.i.d. mixture model.
2. $p(\mathbf{X} \mid \boldsymbol{\theta}) = \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$ naively costs $O(K^N)$ and has no closed-form maximizer, so we use **EM**.
3. $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$ needs only $\gamma(\mathbf{z}_n) = p(\mathbf{z}_n \mid \mathbf{X})$ and $\xi(\mathbf{z}_{n-1}, \mathbf{z}_n) = p(\mathbf{z}_{n-1}, \mathbf{z}_n \mid \mathbf{X})$. The M-step: $\pi_k \propto \gamma(z_{1k})$, $A_{jk} \propto \sum_n \xi(z_{n-1,j}, z_{nk})$, and γ-weighted Gaussian or multinomial emission updates. Never initialize $\boldsymbol{\pi}$ or $\mathbf{A}$ with zeros.
4. **Forward-backward:** $\alpha(\mathbf{z}_n) = p(\mathbf{x}_n|\mathbf{z}_n) \sum_{\mathbf{z}_{n-1}} \alpha(\mathbf{z}_{n-1}) p(\mathbf{z}_n|\mathbf{z}_{n-1})$ from $\alpha(\mathbf{z}_1) = p(\mathbf{z}_1)p(\mathbf{x}_1|\mathbf{z}_1)$; $\beta(\mathbf{z}_n) = \sum_{\mathbf{z}_{n+1}} \beta(\mathbf{z}_{n+1}) p(\mathbf{x}_{n+1}|\mathbf{z}_{n+1}) p(\mathbf{z}_{n+1}|\mathbf{z}_n)$ from $\beta(\mathbf{z}_N) = 1$. Then $\gamma = \alpha\beta / p(\mathbf{X})$, $\xi = \alpha(\mathbf{z}_{n-1}) p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{z}_{n-1}) \beta(\mathbf{z}_n) / p(\mathbf{X})$, $p(\mathbf{X}) = \sum_{\mathbf{z}_N} \alpha(\mathbf{z}_N)$. Cost $O(K^2 N)$. With EM: **Baum-Welch**.
5. The recursions are exactly **sum-product** on the HMM factor graph after absorbing emissions into $f_n(\mathbf{z}_{n-1}, \mathbf{z}_n) = p(\mathbf{z}_n|\mathbf{z}_{n-1}) p(\mathbf{x}_n|\mathbf{z}_n)$.
6. **Scaling:** α underflows; use $\widehat{\alpha}(\mathbf{z}_n) = p(\mathbf{z}_n|\mathbf{x}_{1:n})$ with $c_n = p(\mathbf{x}_n|\mathbf{x}_{1:n-1})$, $c_n \widehat{\alpha}(\mathbf{z}_n) = p(\mathbf{x}_n|\mathbf{z}_n) \sum \widehat{\alpha}(\mathbf{z}_{n-1}) p(\mathbf{z}_n|\mathbf{z}_{n-1})$, and $c_{n+1} \widehat{\beta}(\mathbf{z}_n) = \sum \widehat{\beta}(\mathbf{z}_{n+1}) p(\mathbf{x}_{n+1}|\mathbf{z}_{n+1}) p(\mathbf{z}_{n+1}|\mathbf{z}_n)$. Then $\gamma = \widehat{\alpha}\widehat{\beta}$, $\xi = c_n^{-1} \widehat{\alpha}(\mathbf{z}_{n-1}) p(\mathbf{x}_n|\mathbf{z}_n) p(\mathbf{z}_n|\mathbf{z}_{n-1}) \widehat{\beta}(\mathbf{z}_n)$, and $\ln p(\mathbf{X}) = \sum_n \ln c_n$.
7. **Viterbi** = max-sum: $\omega(\mathbf{z}_{n+1}) = \ln p(\mathbf{x}_{n+1}|\mathbf{z}_{n+1}) + \max_{\mathbf{z}_n} \{\ln p(\mathbf{z}_{n+1}|\mathbf{z}_n) + \omega(\mathbf{z}_n)\}$, $\max \ln p(\mathbf{X}, \mathbf{Z}) = \max \omega(\mathbf{z}_N)$; recover the path by back-tracking $k_n^{\max} = \psi(k_{n+1}^{\max})$. No scaling is needed. The most probable **sequence** can differ from the sequence of individually most probable states.

Acknowledgement: Slide content adapted from C. Bishop, "Pattern Recognition and Machine Learning" (Ch. 13); K. Murphy, "Machine Learning: A Probabilistic Perspective" (Ch. 17); M. I. Jordan, "An Introduction to Probabilistic Graphical Models" (Ch. 12); S. Prince, "Computer Vision: Models, Learning, and Inference" (Ch. 11).